# Predicting Chronic Kidney Disease



PG Group 13

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# 1 Introduction

**TODO:** Motivation, problem, research questions and hypotheses.

# 2 Related Work

**TODO:** Past CKD studies, why our methods, research gap.

# 3 Dataset Description and Analysis

## *3.1 Dataset Overview*

**TODO:** DS1 and image data: source, license, features, statistics, peculiarities, why chosen.

### *3.2 Data Cleaning and Preprocessing*

**TODO:** Steps and why, fit on train only.

### *3.3 Feature Engineering and Selection*

**TODO:** New features, selection method, selected features.

### *3.4 Clustering*

**TODO:** Methods, choice of k, results vs labels.

# 4 Experimental Setup

## *4.1 Evaluation Protocol*

**TODO:** Split, cross-validation, metrics.

## *4.2 Classical Models*

### *4.2.1 k-Nearest Neighbours*

**TODO:** k, distance, why scaling matters.

### *4.2.2 Naive Bayes*

**TODO:** Variant and its assumptions.

### *4.2.3 Random Forest*

**TODO:** Trees, depth, feature importance.

### *4.3 Neural Networks*

#### *4.3.1 Multilayer Perceptron (MLP)*

**TODO:** Layers, activation, optimiser, early stopping.

#### *4.3.2 Convolutional Neural Network (CNN)*

**TODO:** Architecture, augmentation, training.

# 5 Results

## *5.1 Model Results*

**TODO:** Scores, confusion matrices, learning curves, all models compared, hypotheses supported or not.

## *5.2 Further Analyses*

**TODO:** Extra experiments and why.

# 6 Discussion

## *6.1 Findings and Research Questions*

**TODO:** Why the results happened.

## *6.2 Relation to Prior Work*

**TODO:** Comparison with literature.

## *6.3 Practical Implications*

**TODO:** Realism, reliability, risks.

## *6.4 Limitations and Future Work*

**TODO:** Limitations and next steps.

# 7 Conclusion

**TODO:** Short summary, no numbers.

## References

**TODO:** One consistent citation style.

## A Member Contributions

**TODO:** One line per member.